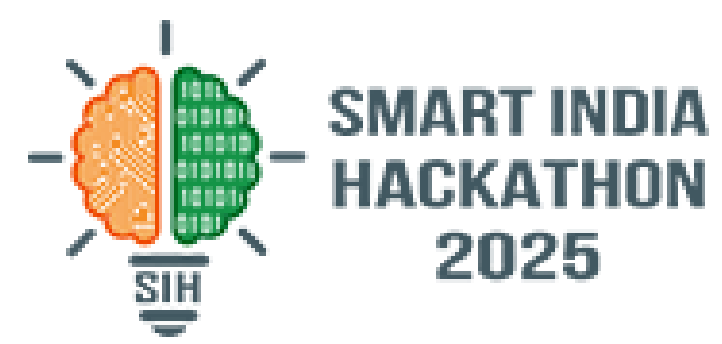


# SMART INDIA HACKATHON 2025



- **Problem Statement ID-SIH25071**
- **Problem Statement Title-AI-Based Rockfall**  
Prediction and Alert System for Open-Pit Mines
- **Theme-** Disaster Management
- **PS Category-** Software
- **Team ID-**
- **Team Name-** 404 Found



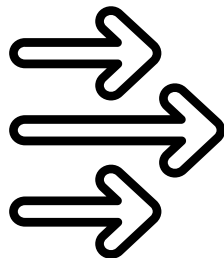
## Problems Addressed

- 1 Traditional system limitations
- 2 Rockfall safety in open-pit mines
- 3 Emergency response & medical response issues
- 4 Worker safety and training deficiencies
- 5 Data Integration and Processing Problems
- 6 Communication and Information Problems

## Proposed solution

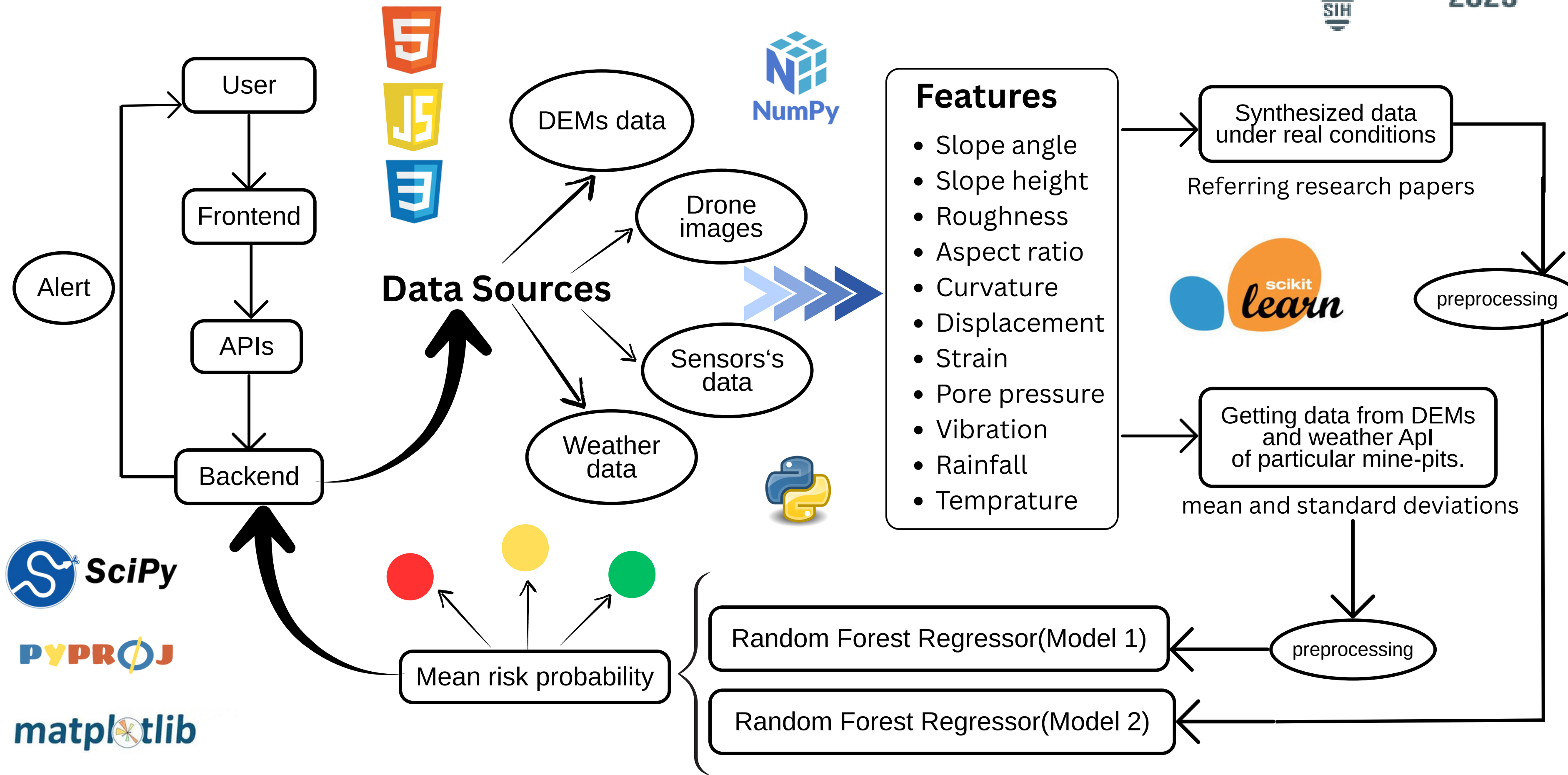
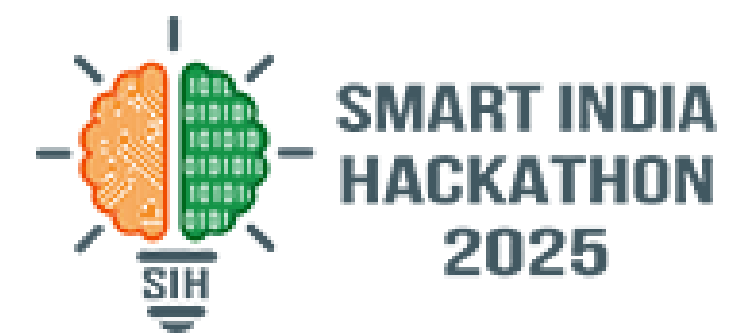
- Emergency Response** - Pre-positioned medical services with real-time worker tracking.
- Smart Alert** - Automated notifications with probability forecasts and safety recommendations
- Interactive Dashboard** - Live risk mapping with vulnerability zones and evacuation guidance
- Worker Safety Integration** - Personal risk assessment with health profiling and training modules
- AI-ML Prediction Engine** - Real-time rockfall risk assessment using multi-source data integration

## Innovation & Uniqueness

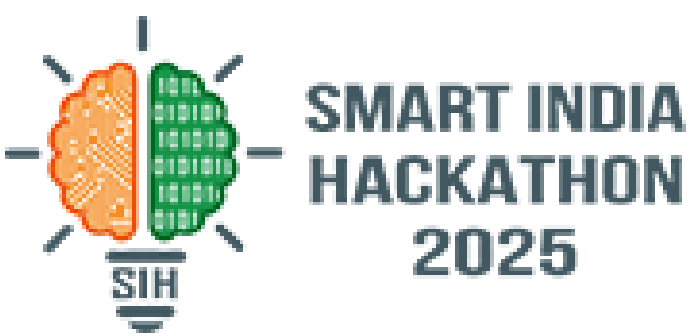


- Personalized Worker Safety Assessment** - Individual health profiling for high-risk zone eligibility
- Predictive Medical Positioning** - AI-driven pre-deployment of medical teams based on risk probabilities
- Integrated Communication Hub** - Multi-modal system with voice alerts, and worker coordination

# TECHNICAL APPROACH



# FEASIBILITY AND VIABILITY



## Feasibility

- **Built with common, ready-to-use tools** for mapping and data analysis
- **Low resource compatibility** crucial for remote mining locations
- **Comprehensive safety features** covering worker health, emergency response, and evacuation
- **Scalable architecture** that can adapt to different mine sizes and types

## Potential Challenges

- **Data Availability & Quality**  
DEM, drone imagery, and sensor data may be incomplete or noisy.
- **Model Training & Accuracy**  
Rockfall events are rare → dataset highly imbalanced.
- **Real-Time Processing**  
Drone & sensor systems generate huge data requiring low-latency processing

## Strategies to overcome

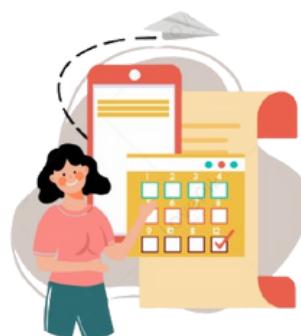
- Synthetic data → simulation tools
- Open-source SRTM(DEM)\_datasets
- Mine history logs integration
- Noise filtering & normalization
- Synthetic event generation
- SMOTE / oversampling
- Ensemble models (RF, XGBoost)
- Cross-validation loops
- GPU acceleration at edge
- Cloud inference (AWS, GCP, Azure)

## Impacts

**Mine Workers & Personnel** - From blind luck to AI-powered guardian angels delivering life-saving personalized alerts



**Mine Management & Planners** - Transform from risky guesswork into crystal ball commanders predicting dangers 24 hours ahead



**Emergency Response Teams** - Evolution from reactive paramedics to "pre-crime" medical units positioned by AI intelligence



**Local Communities** - From living in rockfall fear to sleeping under AI sentinel protection with early disaster warnings



Prevents accidents, boosts worker confidence, and enhances community safety standards

### Social

Prevents ecosystem disruption and promotes sustainable mining practices

### Enviromental

## Benefits

Advances AI safety solutions and establishes predictive safety benchmarks

### Technological

Streamlines safety processes and enables proactive resource management

### Operational

Reduces equipment damage costs, insurance claims, downtime, and legal liabilities

### Economic

# RESEARCH AND REFERENCES

- Review of techniques for monitoring slope instability in open pit mines- [link](#)
- Extensometer monitoring of landslides and slopes.- [link](#)
- Rainfall thresholds for the initiation of landslides, Meteorology and Atmospheric Physics- [link](#)
- Automated identification of rockfalls and landslides using seismic data, Journal of Geophysical Research-[link](#)
- Prediction of rockfall hazard in open pit mines using a regression based machine learning model: [link](#)